



Dynamics and Control (UESTC 3001)



Notes prepared by: Dr Ola R. Popoola



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- Lecture 1: Root Locus Analysis
- Lecture 2: Root Locus II and Nyquist Plots
- **Lecture 3: Bode Plots**
- Lecture 4: Bode Plots II
- Lecture 5: Stability in Frequency Domain
- Lecture 6: Stability Examples
- Lecture 7: Compensators
- Lecture 8: Tutorials and Test Exercises

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Notes prepared by: Dr Ola R. Popoola



Intended Learning Objectives

At the end of this lecture, you will be able to:

- Sketch Bode magnitude plots and Bode phase plots
- Analyze bode plots and compute stability margins



Additional Reading Materials

- “Modern Control Systems” - R. C. Dorf, R.H. Bishop Addison Wesley
- “Modern Control Engineering” - K. Ogata. Prentice Hall International
- “Feedback Control of Dynamic Systems” - Franklin, Powell, Emami-Naeini. Addison Wesley.
- “Feedback Systems: An Introduction for Scientists and Engineers” – K. J. Åström, R. M. Murray. Princeton University Press, Princeton.

Available at

https://www.cds.caltech.edu/~murray/amwiki/index.php/Main_Page

- “Control Engineering: An introduction with the use of Matlab”, D. P. Atherton.

Available at <http://bookboon.com/en/control-engineering-matlab-ebook>

These are recommended only and are not required to pass the course.



Additional Notes

- All required notes/tutorials/exam answers may be found on Moodle.
- You are encouraged to do your own reading to complement the lecture material.
- You can also use the forum on Moodle to ask questions about course material, rather than using email.



Revision

- Root Locus Plots
- Frequency Response Analysis
- Nyquist Plots

Nyquist Plot –Simple Lead

Transfer function: $G(s) = s + 1$

Frequency response function: $G(j\omega) = 1 + j\omega$

This can be seen to have a phase of $\tan^{-1}\omega$. This varies between **0** when $\omega = 0$ and **+90°** when $\omega \rightarrow \infty$.

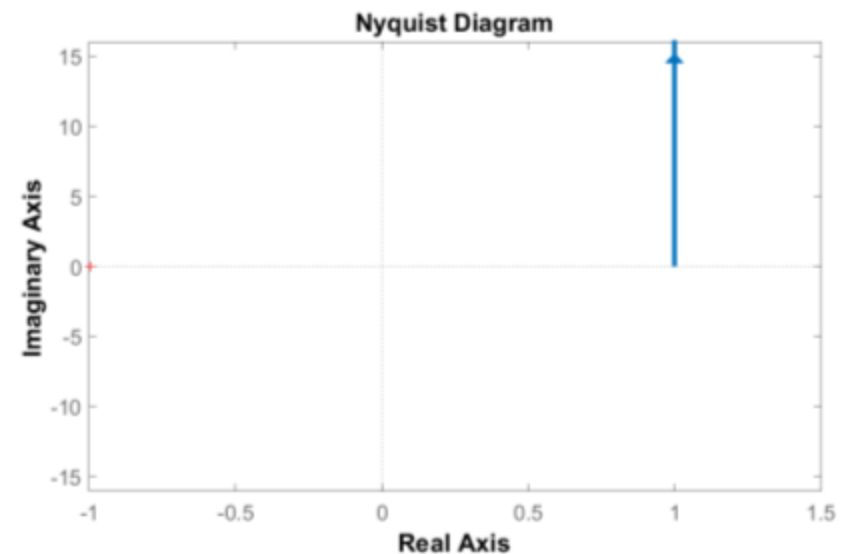
The magnitude given by:

$$|G(j\omega)| = \sqrt{1 + \omega^2}$$

Which varies between

1 when $\omega = 0$ and

$+\infty$ when $\omega \rightarrow \infty$.



Nyquist Plot –MATLAB Examples

$$G(s) = \frac{10(1 + 10s)}{s \left(1 + \frac{s}{10}\right)^2} = \frac{100s + 10}{\frac{1}{100}s^3 + \frac{1}{5}s^2 + s}$$

In Matlab, it can be represented as a transfer function object:

```
sys = tf([100 10],[1/100 1/5 1 0]);
```

- Root locus: `rlocus(sys)`
- Nyquist plot: `nyquist(sys)`

N.B. : don't forget to switch off the display of negative frequencies

Nyquist Plot –Class Examples

Draw the Nyquist plot of the transfer functions:

$$H(s) = \frac{10}{s(s + 2)}$$

- Write out $H(j\omega)$
- Write in complex form i.e. $a + jb$
- Evaluate magnitude and phase as $\omega \rightarrow 0$, $\omega = 2$, and $\omega \rightarrow \infty$
- Sketch Nyquist plot

Nyquist Plot –Class Examples

Draw the Nyquist plot of the transfer functions:

$$H(s) = \frac{10}{s(s+2)}$$

- Write out $H(j\omega)$

$$H(j\omega) = \frac{10}{j\omega(\omega+2)}$$

- Write in complex form i.e. $a + jb$

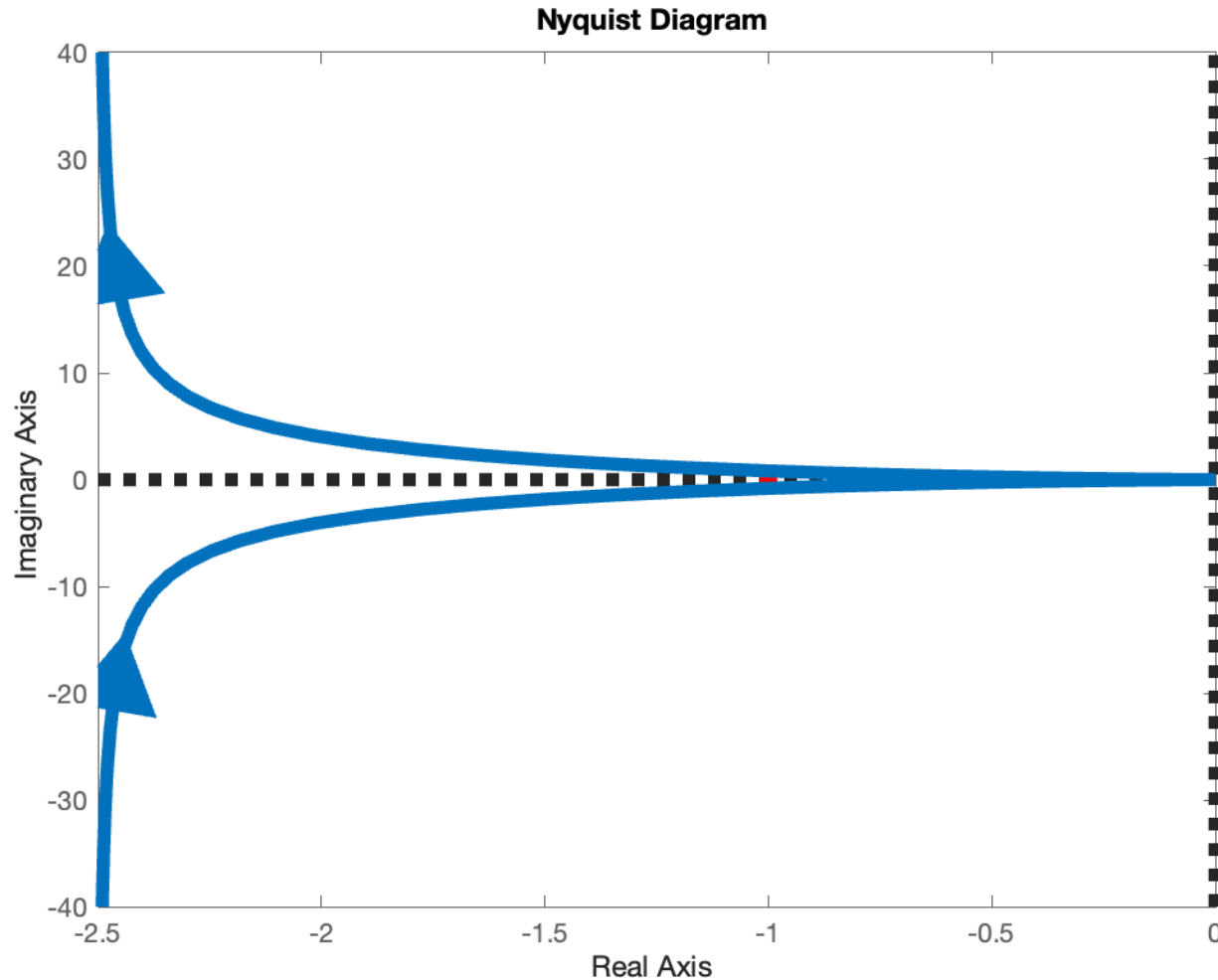
$$H(j\omega) = -\frac{10\omega^2}{\omega^4 + 4\omega^2} - \frac{j20\omega}{\omega^4 + 4\omega^2}$$

- Evaluate magnitude and phase as $\omega \rightarrow 0$, $\omega = 2$, and $\omega \rightarrow \infty$
 - Mag: $-2.5 + j\infty$, $-1.25 - j1.25$, $-0 - j0$
- Sketch Nyquist plot

Nyquist Plot –Class Examples

Draw the Nyquist plot of the transfer functions:

$$(a) H(s) = \frac{10}{s(s+2)}$$



Nyquist Plot –Classwork

Draw the Nyquist plot of the transfer functions:

$$H(s) = \frac{1}{(s - 3)(s - 4)}$$

- Write out $H(j\omega)$
- Write in complex form i.e. $a + jb$
- Evaluate magnitude and phase as $\omega \rightarrow 0$, $\omega = 2$, and $\omega \rightarrow \infty$
- Sketch Nyquist plot



Nyquist Plot –Classwork

Draw the Nyquist plot of the transfer functions:

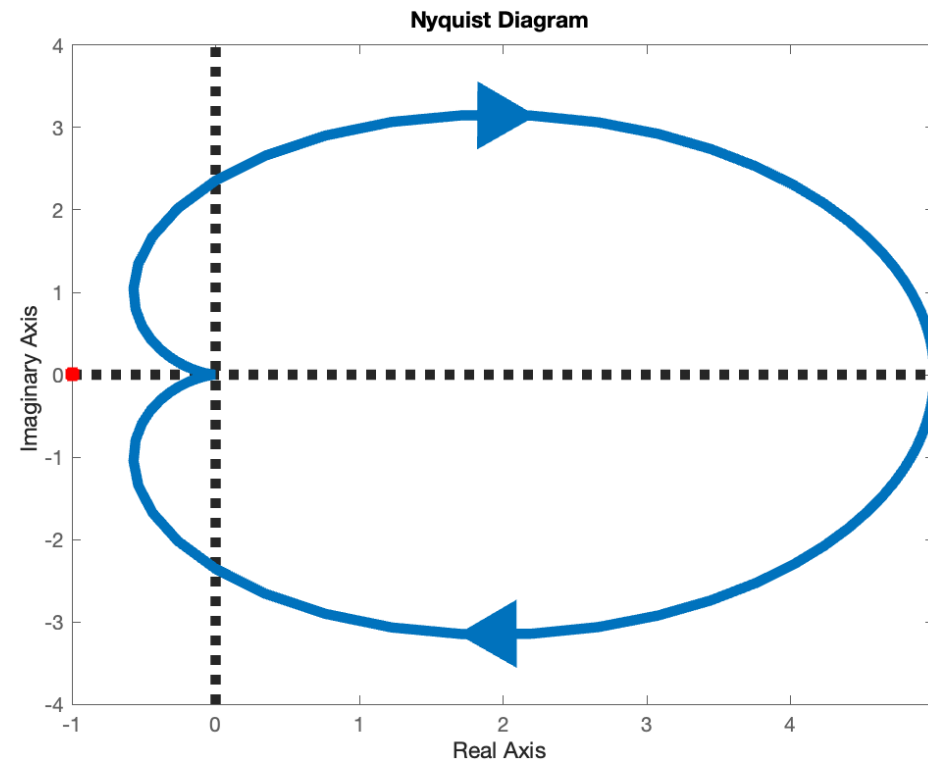
$$H(s) = \frac{1}{(s - 3)(s - 4)}$$

$$H(j\omega) = \frac{12 - \omega^2}{(12 - \omega^2)^2 + (7\omega)^2} + \frac{j7\omega}{(12 - \omega^2)^2 + (7\omega)^2}$$

Nyquist Plot –Classwork

Draw the Nyquist plot of the transfer functions:

$$H(s) = \frac{90}{(s + 3)(s + 6)}$$





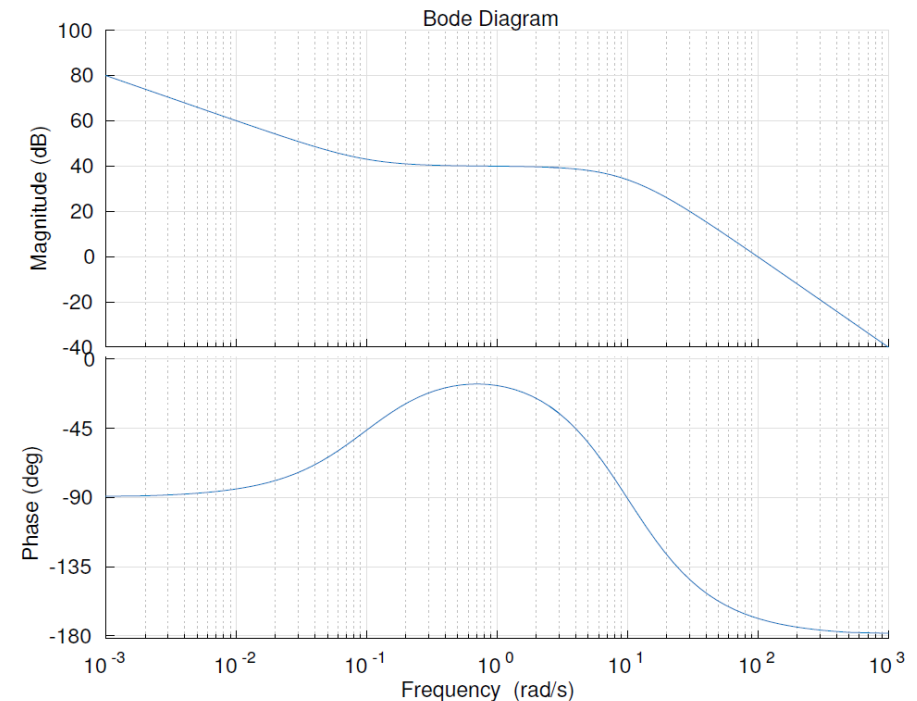
Bode Plots/Diagrams

Bode Plots Introduction

Bode Plots are an alternative method for displaying frequency responses graphically.

The Bode plots use two separate plots

- Magnitude/gain
- Phase



Why Bode Plots?

- Nyquist plots for complex systems are hard to draw accurately without reasonable computations.
- Bode plots are sometimes used instead as accurate sketches are possible.
- To guide us to draw the Bode diagrams of ‘complex’ transfer functions, we can use asymptotic plots.

Bode Components and Time Constant form

- The general form a transfer function can be factorised by writing in the time constant form such as:

$$G(s) = \frac{K \left(1 + \frac{s}{\omega_{z1}}\right) \left(1 + \frac{s}{\omega_{z2}}\right) \dots}{s^r \left(1 + \frac{s}{\omega_{p1}}\right) \left(1 + \frac{s}{\omega_{p2}}\right) \dots}$$

- The plots are:

gain measured in dB = $20 \log_{10} |G(j\omega)|$

Phase, $\arg[G(j\omega)] = \angle G(j\omega)$

Plotted against frequency, ω , on a logarithmic scale.



Writing transfer functions in time constant form

Time constant form –Classwork

Write the transfer functions below in the time constant form:

$$H(s) = \frac{1}{s(s + 2)}$$

$$H(s) = \frac{1}{(s + 3)(s + 4)}$$

$$H(s) = \frac{1}{s^2 + 3s + 3}$$

For quadratic polynomials, comment on transfer function and damping factor.

Bode Plots -Magnitude

Magnitude:

$$G(s) = \frac{K \left(1 + \frac{s}{\omega_{z1}}\right) \left(1 + \frac{s}{\omega_{z2}}\right) \dots}{s^r \left(1 + \frac{s}{\omega_{p1}}\right) \left(1 + \frac{s}{\omega_{p2}}\right) \dots}$$

$$\begin{aligned} 20 \log_{10} |G(j\omega)| &= 20 \log_{10} K \\ &+ 20 \log_{10} \left| 1 + j \frac{\omega}{\omega_{z1}} \right| + 20 \log_{10} \left| 1 + j \frac{\omega}{\omega_{z2}} \right| \\ &- 20r \log_{10} |j\omega| \\ &- 20 \log_{10} \left| 1 + j \frac{\omega}{\omega_{p1}} \right| - 20 \log_{10} \left| 1 + j \frac{\omega}{\omega_{p2}} \right| \end{aligned}$$

Code Plots -Phase

$$G(s) = \frac{K \left(1 + \frac{s}{\omega_{z1}}\right) \left(1 + \frac{s}{\omega_{z2}}\right) \dots}{s^r \left(1 + \frac{s}{\omega_{p1}}\right) \left(1 + \frac{s}{\omega_{p2}}\right) \dots}$$

Phase:

$$\begin{aligned} \angle G(j\omega) = \text{Arg}[G(j\omega)] &= \tan^{-1} \left(\frac{\omega}{\omega_{z1}} \right) + \tan^{-1} \left(\frac{\omega}{\omega_{z2}} \right) + \dots \\ &\quad - r \cdot 90^\circ \\ &\quad - \tan^{-1} \left(\frac{\omega}{\omega_{p1}} \right) - \tan^{-1} \left(\frac{\omega}{\omega_{p2}} \right) + \dots \end{aligned}$$

Bode Plots –Simple Gain

Simple Gain:

$$G(s) = \frac{K \left(1 + \frac{s}{\omega_{z1}}\right) \left(1 + \frac{s}{\omega_{z2}}\right) \dots}{s^r \left(1 + \frac{s}{\omega_{p1}}\right) \left(1 + \frac{s}{\omega_{p2}}\right) \dots}$$

$$G(s) = K$$

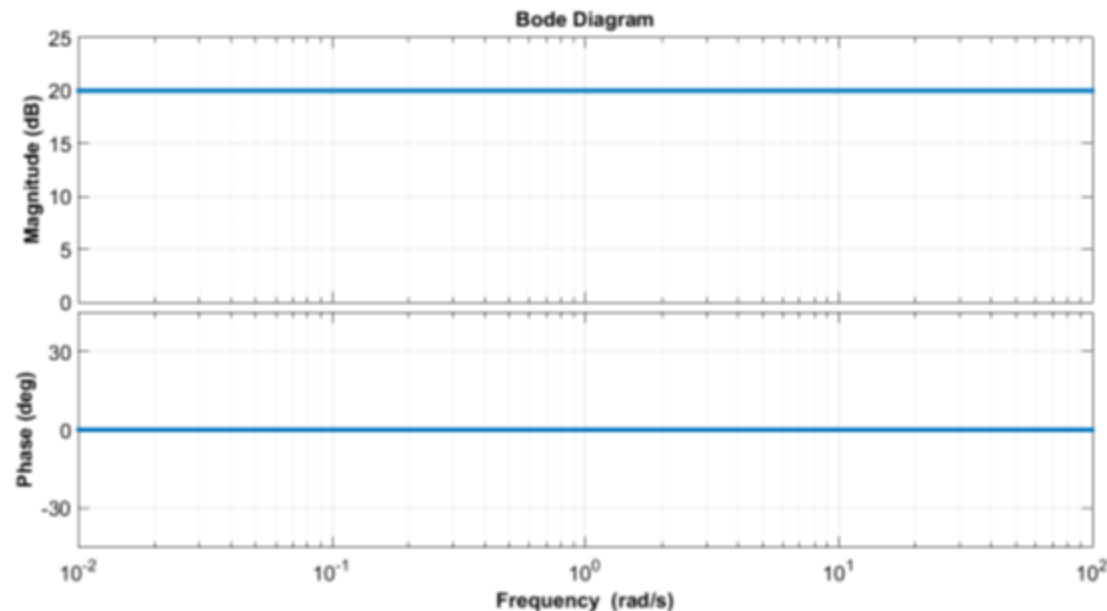
Gain:

$$20\log_{10}|G| = 20\log_{10}|K|$$

Phase:

$$\arg G(j\omega) = 0^\circ$$

Example for K=10



Bode Plots -Integrator

Integrator:

$$G(s) = \frac{K \left(1 + \frac{s}{\omega_{z1}}\right) \left(1 + \frac{s}{\omega_{z2}}\right) \dots}{s^r \left(1 + \frac{s}{\omega_{p1}}\right) \left(1 + \frac{s}{\omega_{p2}}\right) \dots}$$

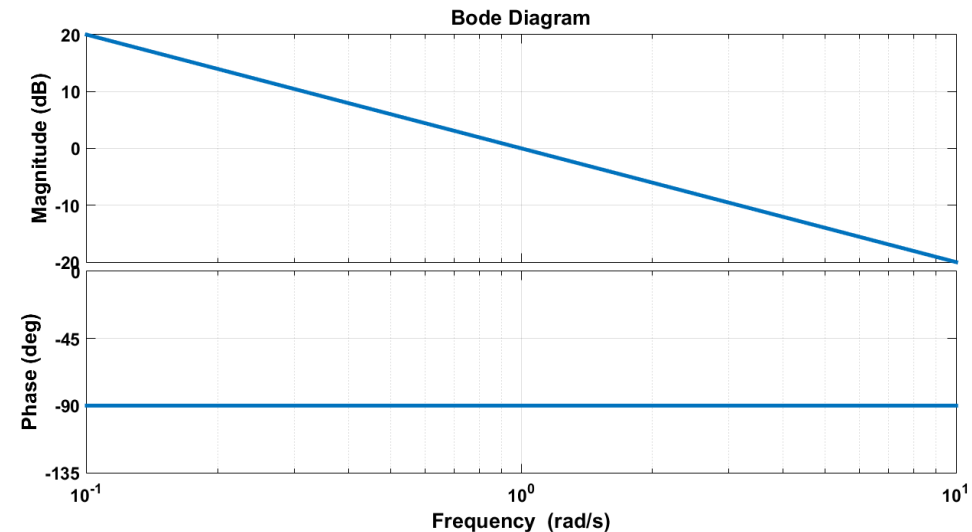
$$G(s) = \frac{1}{s}$$

Gain:

$$20\log_{10}|G| = -20\log_{10}\omega$$

Phase:

$$\arg G(j\omega) = -90^\circ$$



Bode Plots –Simple lag Gain

• **Simple lag:** $G(s) = \frac{1}{1 + \frac{s}{\omega_c}}$

$$G(s) = \frac{K \left(1 + \frac{s}{\omega_{z1}}\right) \left(1 + \frac{s}{\omega_{z2}}\right) \dots}{s^r \left(1 + \frac{s}{\omega_{p1}}\right) \left(1 + \frac{s}{\omega_{p2}}\right) \dots}$$

Gain:

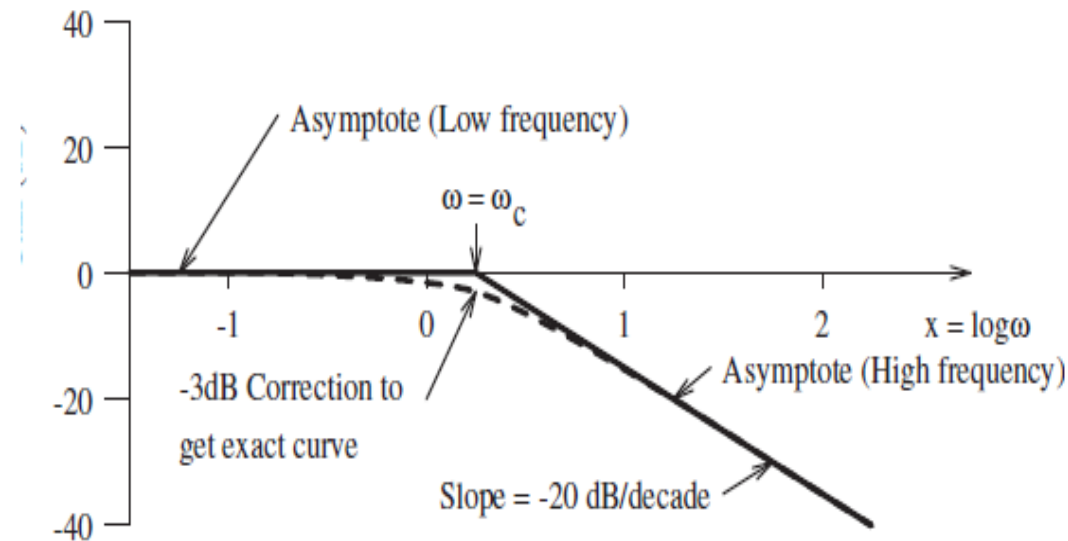
$$20\log_{10}|G| = -10\log_{10} \left(1 + \frac{\omega^2}{\omega_c^2}\right)$$

$\omega \ll \omega_c \rightarrow -10\log_{10}(1) = 0 \text{ dB}$

$\omega = \omega_c \rightarrow -10\log_{10}(2) = -3 \text{ dB}$

$\omega \gg \omega_c \rightarrow -20\log_{10} \left(\frac{\omega}{\omega_c}\right) \rightarrow -20 \text{ dB/decade crossing the 0 dB line at}$

$\omega = \omega_c$



Bode Plots –Simple Lag Phase

- Simple lag:**

$$G(s) = \frac{1}{1 + \frac{s}{\omega_c}}$$

$$G(s) = \frac{K \left(1 + \frac{s}{\omega_{z1}}\right) \left(1 + \frac{s}{\omega_{z2}}\right) \dots}{s^r \left(1 + \frac{s}{\omega_{p1}}\right) \left(1 + \frac{s}{\omega_{p2}}\right) \dots}$$

Phase: $\tan^{-1} G = -\tan^{-1} \frac{\omega}{\omega_c}$

$$\omega \ll \omega_c \rightarrow -\tan^{-1} \frac{\omega}{\omega_c} \simeq 0^\circ$$

$$\omega = \omega_c \rightarrow -\tan^{-1} \frac{\omega_c}{\omega_c} = -45^\circ$$

$$\omega \gg \omega_c \rightarrow -\tan^{-1} \frac{\omega}{\omega_c} \simeq -90^\circ$$

What is an acceptable limit for $\omega \ll \omega_c$ (resp. $\omega \gg \omega_c$) ?

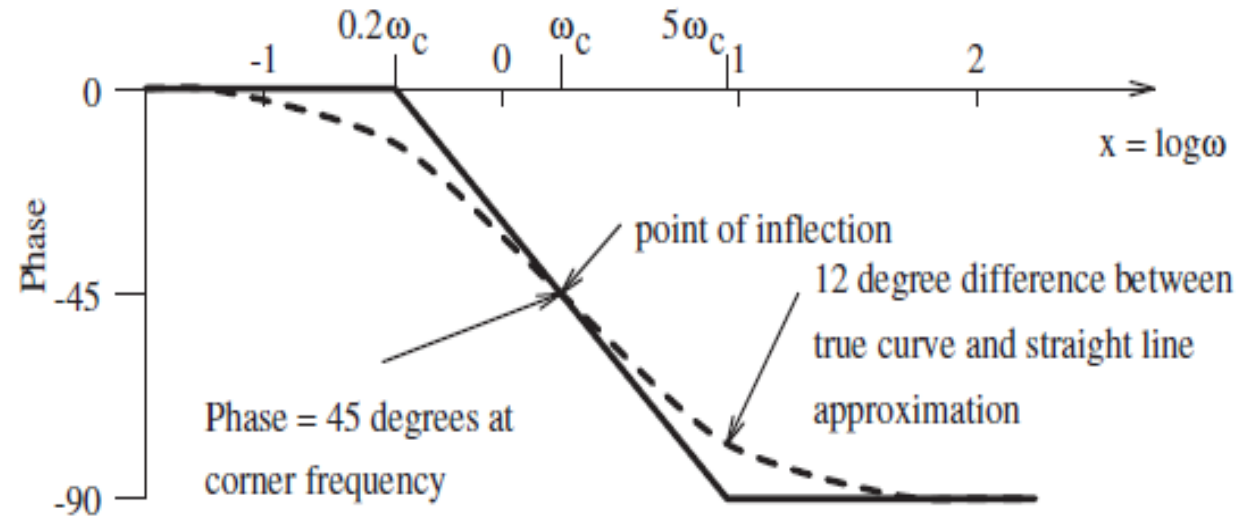
$$\frac{\omega_c}{5} \text{ (resp. } 5\omega_c)$$

or

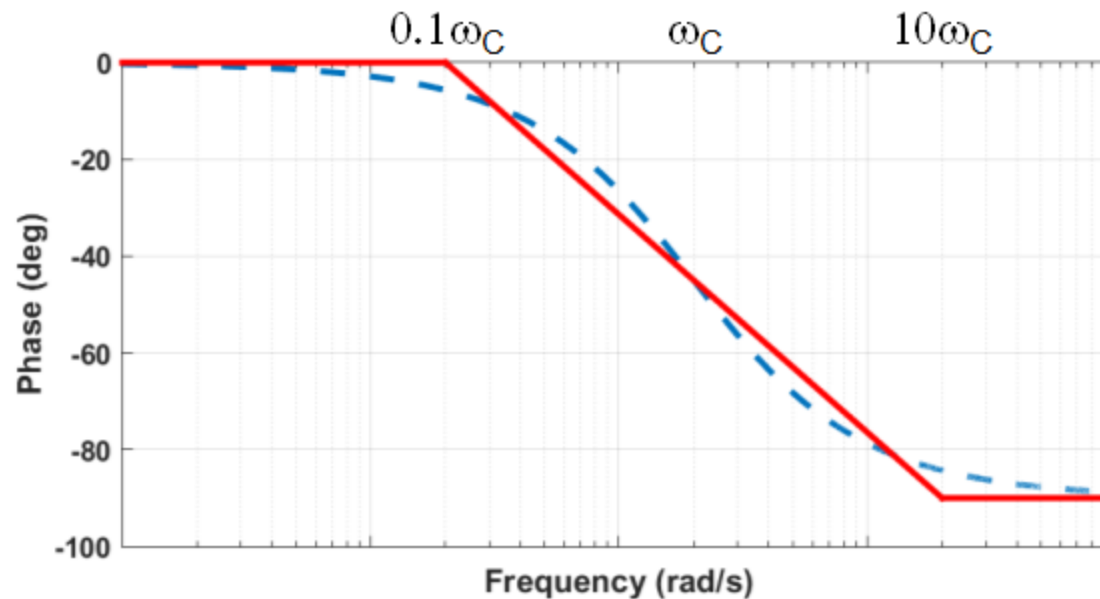
$$\frac{\omega_c}{10} \text{ (resp. } 10\omega_c)$$

Bode Plots – Simple lag phase plot

Version 1



Version 2



Bode Plots –Simple Lead Gain

- Simple lead $G(s) = 1 + \frac{s}{\omega_c}$

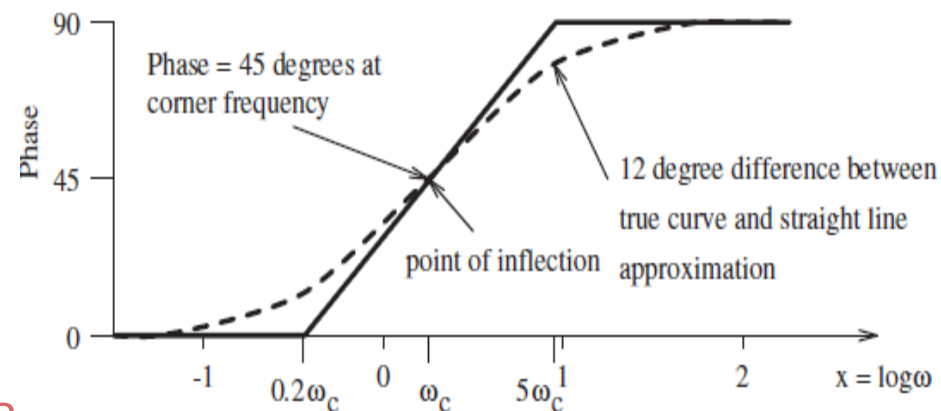
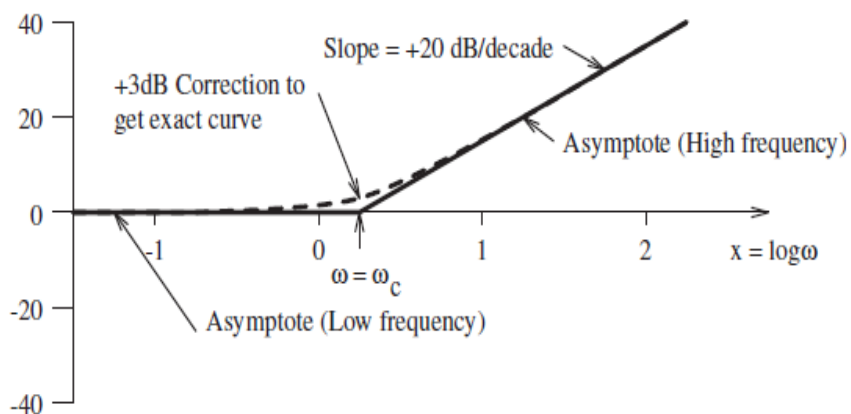
$$G(s) = \frac{K \left(1 + \frac{s}{\omega_{z1}}\right) \left(1 + \frac{s}{\omega_{z2}}\right) \dots}{s^r \left(1 + \frac{s}{\omega_{p1}}\right) \left(1 + \frac{s}{\omega_{p2}}\right) \dots}$$

Gain: $20\log_{10}|G| = 10\log_{10} \left(1 + \frac{\omega^2}{\omega_c^2}\right)$

$\omega \ll \omega_c \rightarrow 10\log_{10}(1) = 0 \text{ dB}$

$\omega = \omega_c \rightarrow 10\log_{10}(2) = 3 \text{ dB}$

$\omega \gg \omega_c \rightarrow 20\log_{10} \left(\frac{\omega}{\omega_c}\right) \rightarrow 20 \text{ dB/decade crossing the } 0 \text{ dB line at } \omega = \omega_c$



Contribution of Components

$$G(s) = \frac{K \left(1 + \frac{s}{\omega_{z1}}\right) \left(a + \frac{s}{\omega_{z2}}\right) \dots}{s^r \left(1 + \frac{s}{\omega_{p1}}\right) \left(1 + \frac{s}{\omega_{p2}}\right) \dots \left(1 + \frac{2\zeta s}{\omega_n} + \frac{s^2}{\omega_n^2}\right) \dots}$$

Component	Formular	Gain	Phase
a. Gain	K	$20 \log K $	0°
b. Integrator	$\frac{1}{s^r}$	$-20r \log \omega$	$-r \cdot 90^\circ$
c. First Order lead	$1 + \frac{s}{\omega_{z1}}$	0, 20 dB/dec after ω_{z1}	$\tan^{-1}\left(\frac{\omega}{\omega_{z1}}\right)$
d. First Order lag	$\frac{1}{1 + \frac{s}{\omega_{p1}}}$	0, -20 dB/dec after ω_{p1}	$-\tan^{-1}\left(\frac{\omega}{\omega_{p1}}\right)$
e. Second Order lag	$\frac{1}{\left(1 + \frac{2\zeta s}{\omega_n} + \frac{s^2}{\omega_n^2}\right)}$	0, -40 dB/dec after $\omega = \omega_n$	$-\tan^{-1}\left(\frac{\frac{2\zeta\omega}{\omega_n}}{1 - \frac{\omega^2}{\omega_n^2}}\right)$



Code Plot Process

Now let us discuss the procedure of drawing a Bode plot:

1. Substitute the $s = j\omega$ in the open loop transfer function $G(s) \times H(s)$.
2. Find the corresponding corner frequencies and tabulate them.
3. Now we are required one semi-log graph chooses a frequency range such that the plot should start with the frequency which is lower than the lowest corner frequency. Mark angular frequencies on the x-axis, mark slopes on the left hand side of the y-axis by marking a zero slope in the middle and on the right hand side mark phase angle by taking -180° in the middle.
4. Calculate the gain factor and the type of order of the system.
5. calculate slope corresponding to each factor then add.

Bode Plots –Examples

$$G(s) = \frac{10(1 + 10s)}{s \left(1 + \frac{s}{10}\right)^2}$$

Components:

A simple gain 10

A simple lead term $(1 + 10s)$

An integrator $\frac{1}{s}$

Two lag terms $\frac{1}{1 + \frac{s}{10}}$

Bode Plots – Examples

$$G(s) = \frac{10(1 + 10s)}{s \left(1 + \frac{s}{10}\right)^2}$$

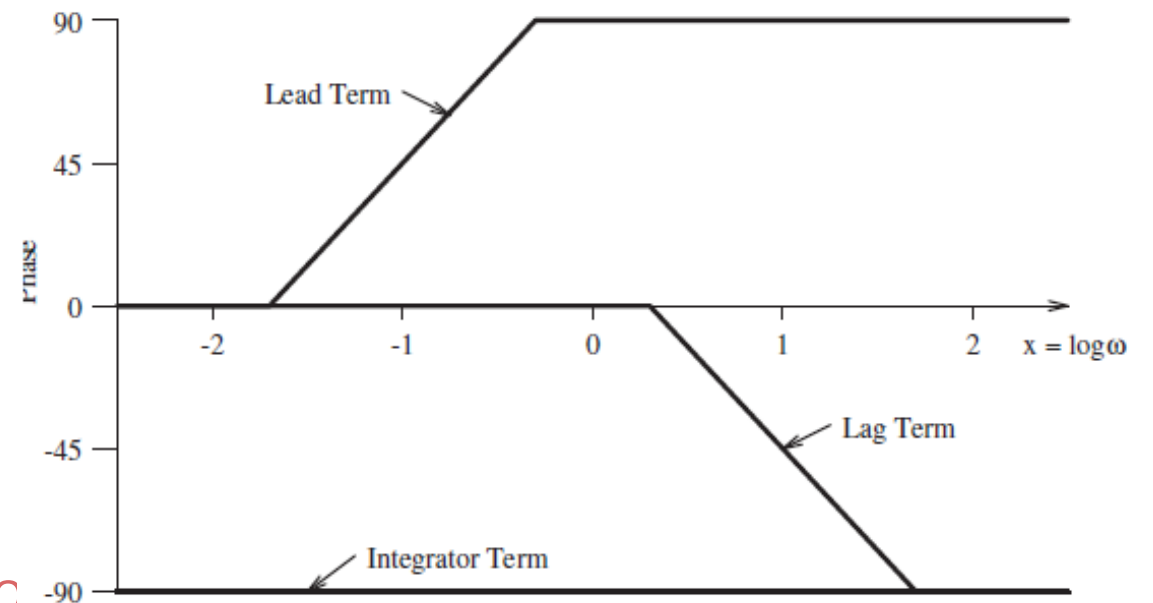
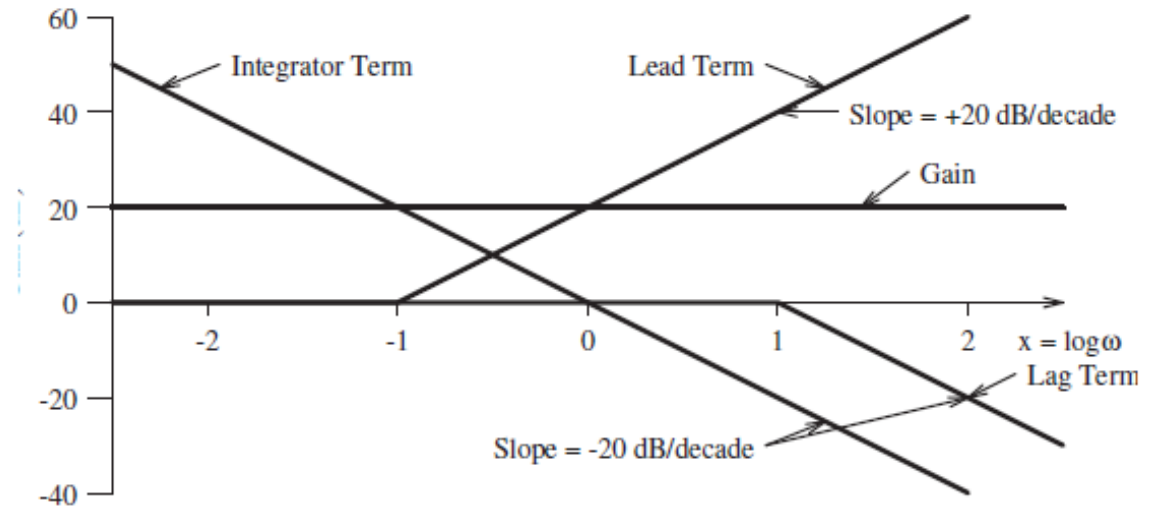
Components:

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An integrator $\frac{1}{s}$

Two lag terms $\frac{1}{1 + \frac{s}{10}}$



Bode Plots – Examples

$$G(s) = \frac{10(1 + 10s)}{s \left(1 + \frac{s}{10}\right)^2}$$

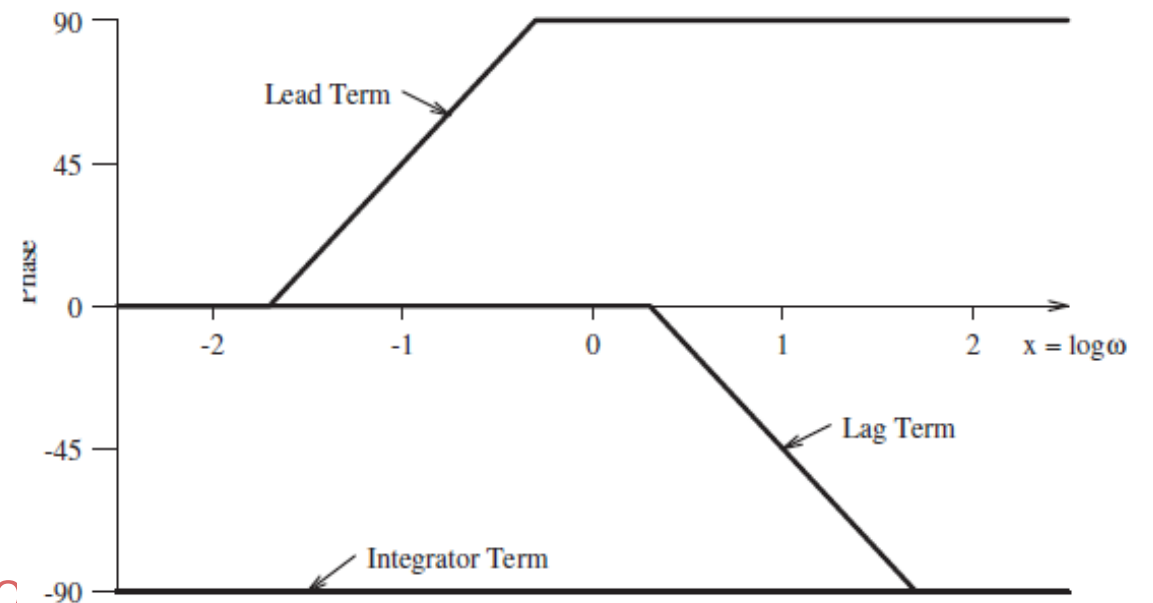
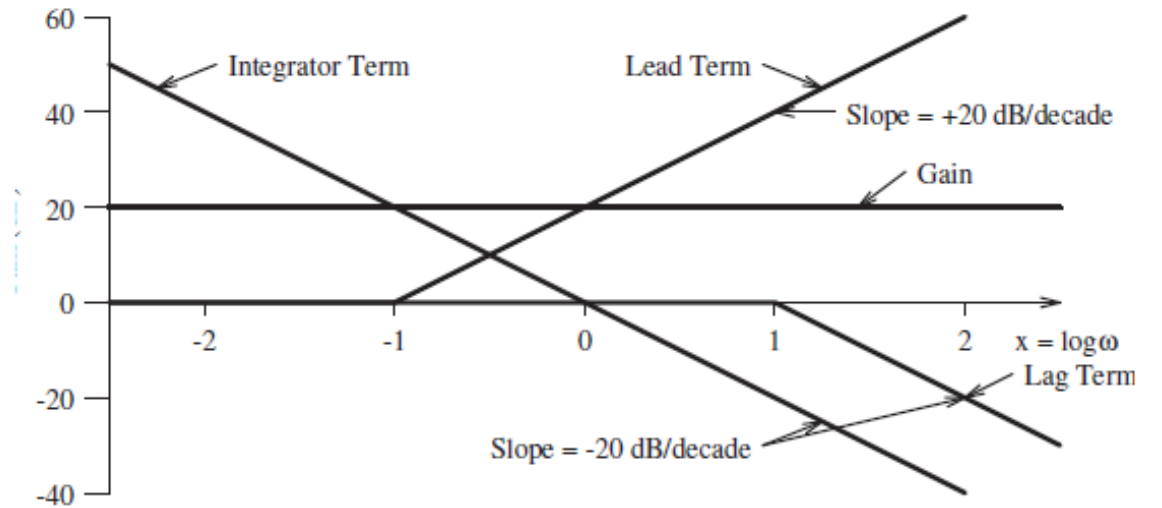
Components:

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An integrator $\frac{1}{s}$

Two lag terms $\frac{1}{1 + \frac{s}{10}}$

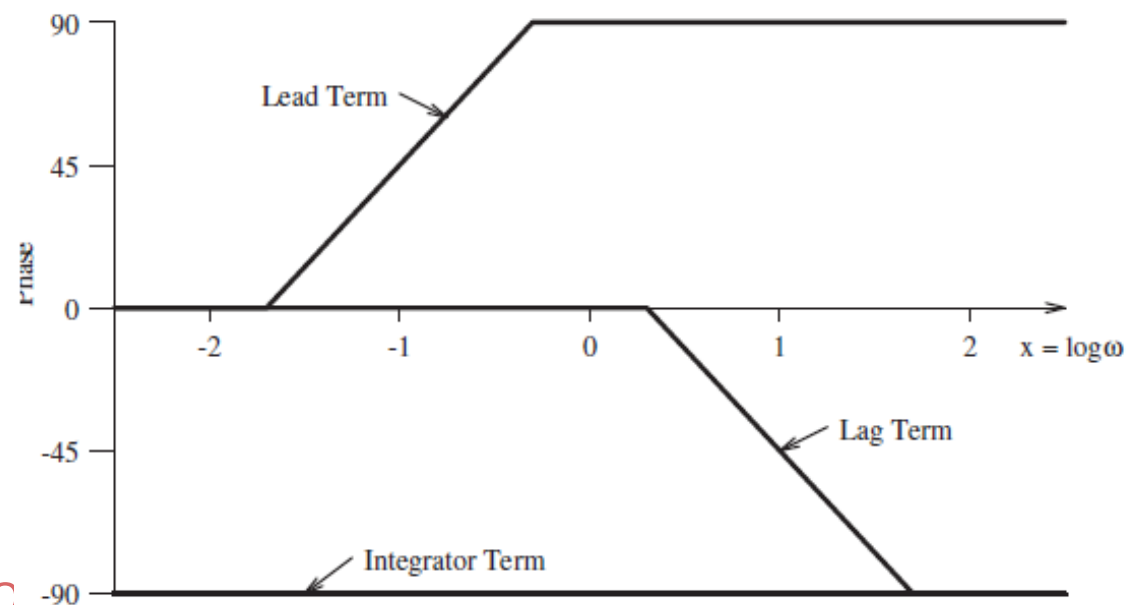
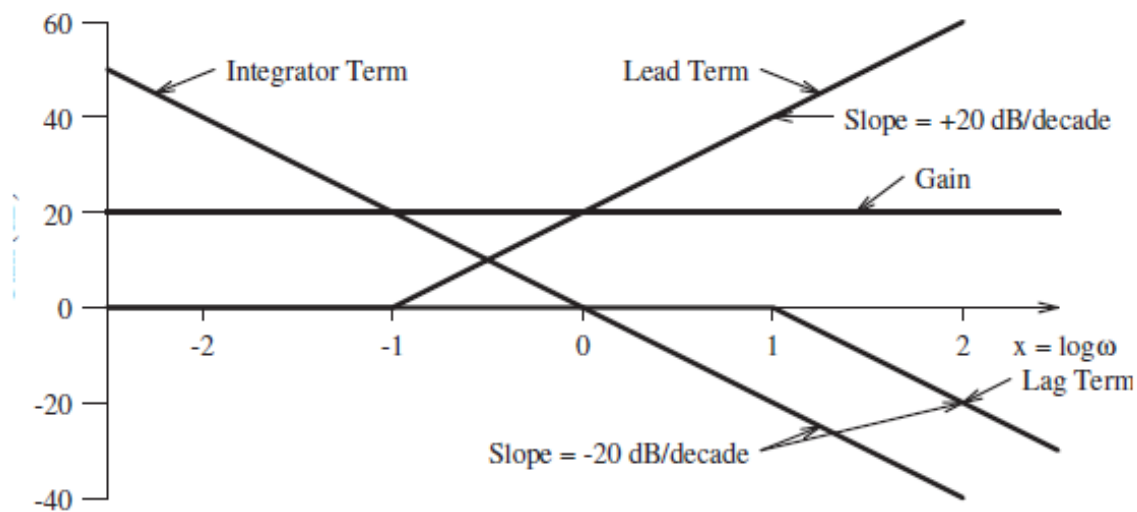


Bode Plots – Examples

$$G(s) = \frac{10(1 + 10s)}{s \left(1 + \frac{s}{10}\right)^2}$$

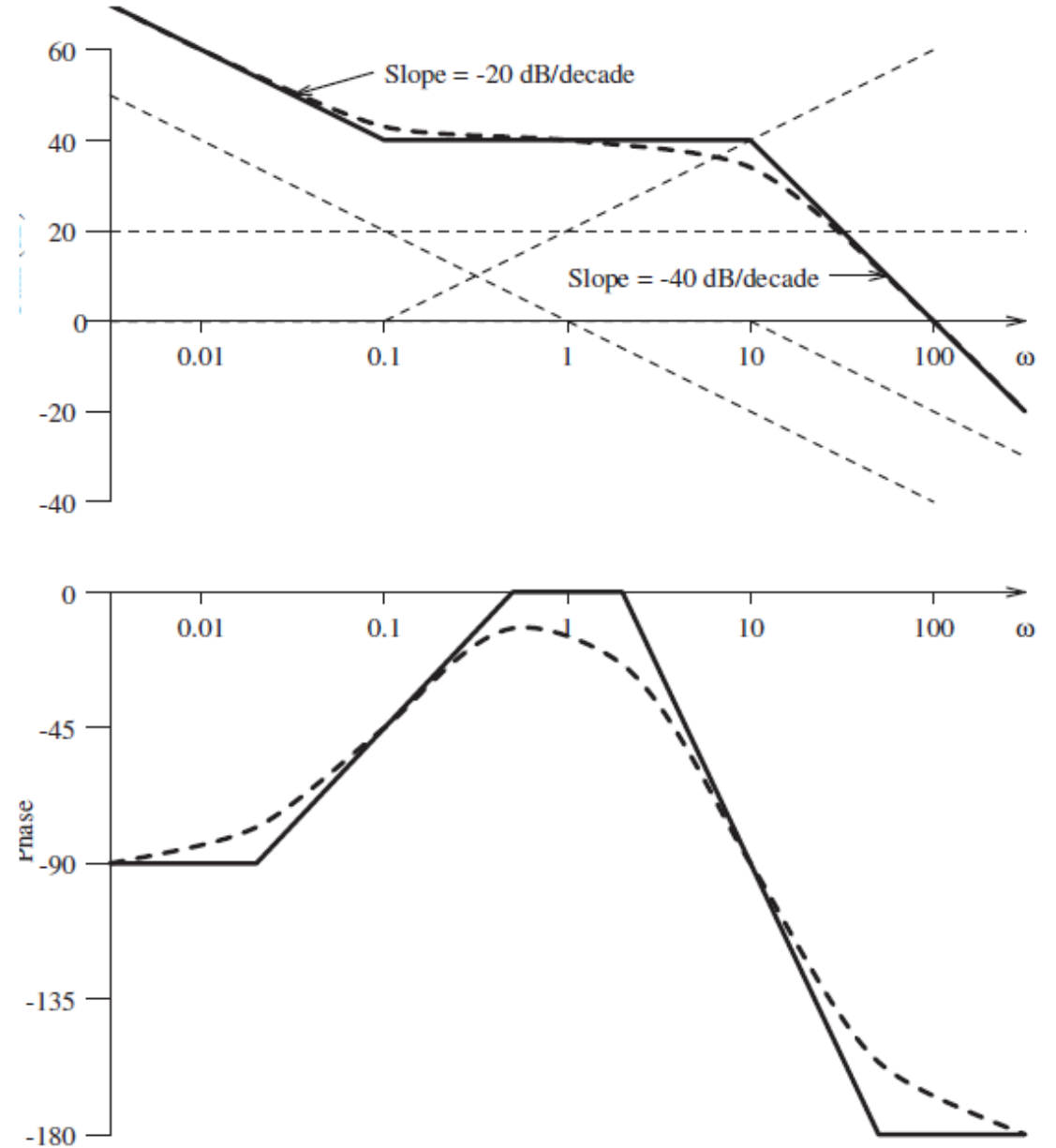
Phase Plot

$0 < \omega < 0.02$	Angle = -90°
$\omega = 0.1$	Angle = -45°
$0.5 < \omega < 2$	Angle = 0°
$\omega = 10$	Angle = -90°
$50 < \omega < \infty$	Angle = -180°



Bode Plots – Examples

$$G(s) = \frac{10(1 + 10s)}{s \left(1 + \frac{s}{10}\right)^2}$$



Bode Plots –Matlab

$$G(s) = \frac{10(1 + 10s)}{s \left(1 + \frac{s}{10}\right)^2} = \frac{100s + 10}{\frac{1}{100}s^3 + \frac{1}{5}s^2 + s}$$

In Matlab, it can be represented as a transfer function object:

`sys = tf([100 10],[1/100 1/5 1 0]);`

Bode plot: `bodeplot(sys)`

Nyquist plot: `nyquistplot(sys)`

Response of a delay

- Laplace transform of a delay of T seconds is:

$$H(s) = e^{-sT}$$

In the frequency domain:

$$s = j\omega$$

The frequency response of a delay is:

$$H(j\omega) = e^{-j\omega T}$$

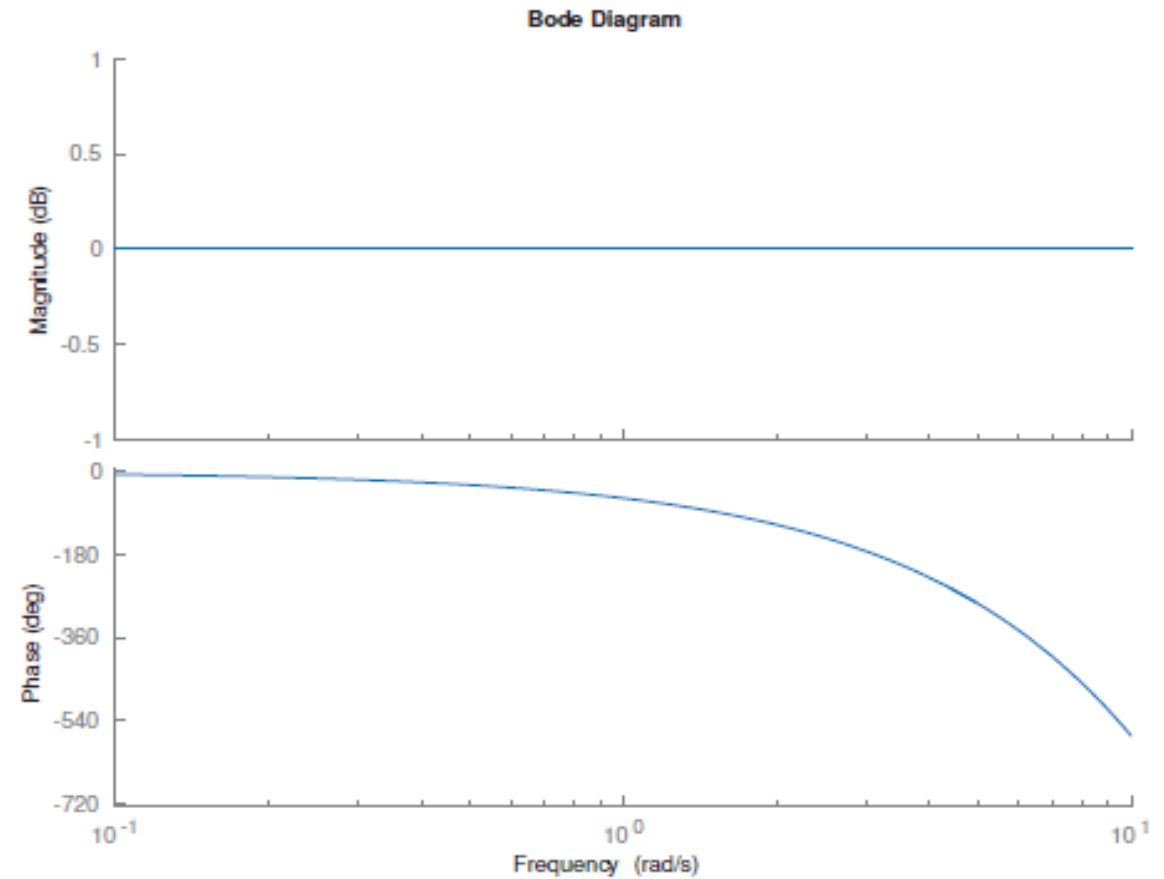
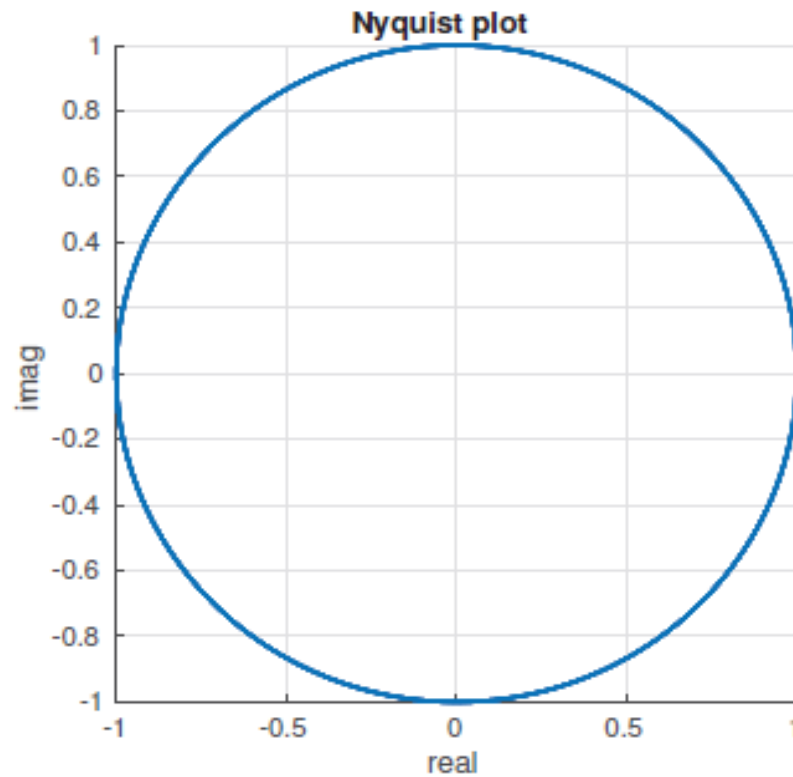
Magnitude:

$$|H(j\omega)| = 1$$

Phase:

$$\angle H(j\omega) = -\omega T \text{ radians}$$

Response of a delay





Practice Exercise

Consider the transfer function:

$$H(s) = \frac{10 \cdot (s + 0.5)}{(s + 10)}$$

For this transfer function, find its Bode diagrams